# Surface Areas and Volumes

## The big idea

Real objects are often **combinations** of basic solids: a capsule is a cylinder with two hemispheres; a toy top is a cone on a hemisphere. We find their surface area by adding only the **visible** surfaces, and their volume by adding the volumes of the parts.

## Formulas you need

| Solid | Curved/Lateral surface | Total surface | Volume |
|---|---|---|---|
| Cuboid (l, b, h) | $2h(l + b)$ | $2(lb + bh + hl)$ | $lbh$ |
| Cube (a) | $4a^2$ | $6a^2$ | $a^3$ |
| Cylinder (r, h) | $2\pi rh$ | $2\pi r(r + h)$ | $\pi r^2 h$ |
| Cone (r, h, l) | $\pi rl$ | $\pi r(l + r)$ | $\frac{1}{3}\pi r^2 h$ |
| Sphere (r) | $4\pi r^2$ | $4\pi r^2$ | $\frac{4}{3}\pi r^3$ |
| Hemisphere (r) | $2\pi r^2$ | $3\pi r^2$ | $\frac{2}{3}\pi r^3$ |

For a cone, the slant height $\mathbf{l = \sqrt{(r^2 + h^2)}}$.

## Surface area of combined solids

**ADD ONLY WHAT YOU CAN TOUCH**

When two solids are joined, the joined faces are hidden. Use **curved** surface areas of the parts, not total surface areas.

**A CAPSULE: CYLINDER WITH A HEMISPHERE AT EACH END. TOTAL LENGTH 14 MM, DIAMETER 5 MM. FIND ITS SURFACE AREA.**

r = 2.5 mm. Cylinder length = 14 − 5 = 9 mm.

Surface = $2 \times (2\pi r^2) + 2\pi rh = 4\pi r^2 + 2\pi rh = 2\pi r(2r + h) = 2 \times (22/7) \times 2.5 \times (5 + 9)$ = **220 mm²**.

## Volume of combined solids

**A SOLID IS A CONE OF RADIUS 1 CM AND HEIGHT 1 CM ON A HEMISPHERE OF RADIUS 1 CM. FIND ITS VOLUME IN TERMS OF Π.**

Volume = $\frac{1}{3}\pi(1)^2(1) + \frac{2}{3}\pi(1)^3 = \frac{1}{3}\pi + \frac{2}{3}\pi =$ **π cm³**.

**VOLUME STAYS, SURFACE CHANGES**

When a solid is melted and recast, **volume is conserved**: volume of the old shape = total volume of the new shapes. The surface area usually changes.

**USING H INSTEAD OF L**

The curved surface of a cone uses the **slant height l**, not the vertical height h.

## Quick recap

- Learn the table of formulas.
- Combined surface area: add curved surfaces only, never the hidden joins.
- Combined volume: add (or subtract) the volumes of the parts.

## Practice questions

1. The curved surface area of a hemisphere of radius r is
   (a) $\pi r^2$ (b) $2\pi r^2$ (c) $3\pi r^2$ (d) $4\pi r^2$
2. A cone has radius 3 cm and height 4 cm. Its slant height is
   (a) 7 cm (b) 5 cm (c) 6 cm (d) 12 cm
3. The volume of a cylinder of radius 7 cm and height 10 cm is ($\pi = 22/7$)
   (a) 440 $cm^3$ (b) 1540 $cm^3$ (c) 154 $cm^3$ (d) 2200 $cm^3$
4. A cone and a cylinder have the same radius and height. The ratio of their volumes is
   (a) 1 : 3 (b) 3 : 1 (c) 1 : 2 (d) 2 : 3
5. When a metal sphere is melted and recast into small spheres, what stays the same?
   (a) Total surface area (b) Total volume (c) Radius (d) Number of spheres

Answers: 1. b 2. b 3. b 4. a 5. b